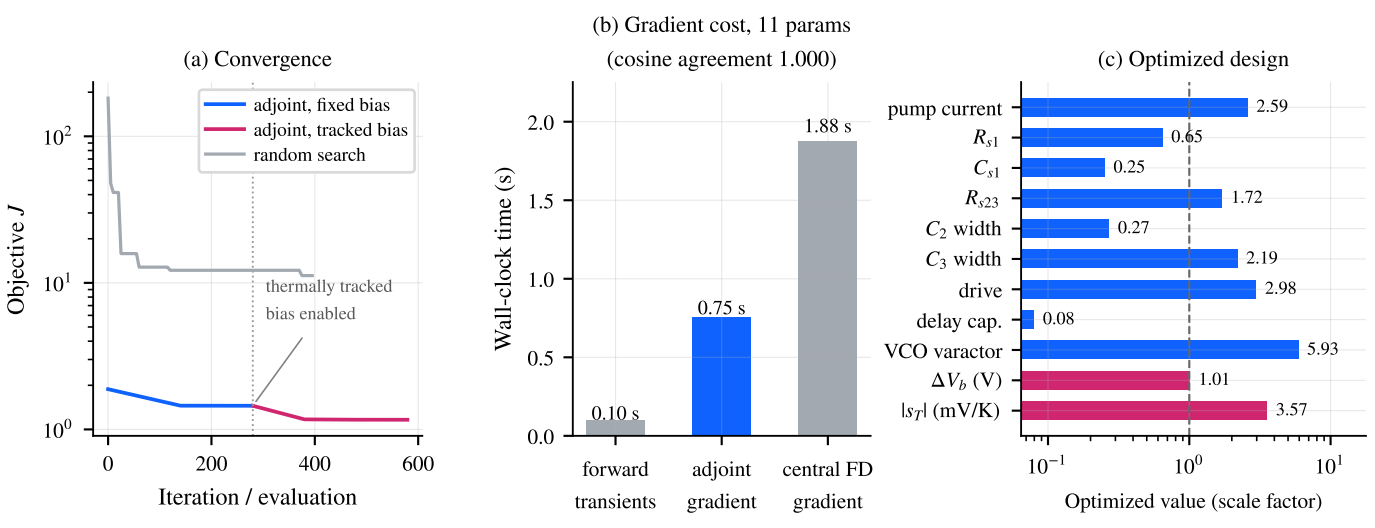
(a) Convergence
adjoint, fixed bias
adjoint, tracked bias
random search
thermally tracked
bias enabled
Objective J
10^2
10^1
10^0
0
200
400
600
Iteration / evaluation
(b) Gradient cost, 11 params
(cosine agreement 1.000)
Wall-clock time (s)
0.0
0.5
1.0
1.5
2.0
0.10 s
0.75 s
1.88 s
forward transients
adjoint gradient
central FD gradient
(c) Optimized design
pump current
R_{s1}
C_{s1}
R_{s23}
C_2 width
C_3 width
drive
delay cap.
VCO varactor
ΔV_b (V)
$|s_T|$ (mV/K)
2.59
0.65
0.25
1.72
0.27
2.19
2.98
0.08
5.93
1.01
3.57
10^{-1}
10^0
10^1
Optimized value (scale factor)